

FAI Governance Health Indicators

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This work derives from and formalizes the inter-Self coordination architecture introduced in “Inter-Self Coordination via Shared Substrate / Full Aspect Integration” (Li, April 2026), the third paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026) and “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026).

Abstract

Phase D2 of the CKS derivation series has established the operational governance record for Full Aspect Integration (FAI) events: configuration as substrate content (D1.22), the conflict registry (D2.13), escalation responsiveness timelines (D2.14), exchange bounding verification (D2.16), home perimeter integrity verification (D2.17), FAI-origin provenance attribution (D2.10), standing configuration currency (D2.21), trust calibration coherence (D2.29), and post-emergency investigation (D2.28). D2.35 synthesizes these records into a formal indicator set: ten positive health indicators and ten warning indicators, each readable from governance records alone without reliance on participant self-reporting or interviews. Together the twenty indicators constitute a governance health dashboard for periodic review of a Self’s FAI governance portfolio. This note formalizes the indicators, maps each warning indicator to the anti-pattern trajectories it precedes, and provides an operational test for the dashboard’s completeness.

1. The record-readability premise

Governance health in the FAI architecture is not a subjective assessment. It is an observable property of the governance records the architecture requires. The Phase D2 operational governance record was designed so that every architectural commitment produces a durable record: configuration records, authorization records, conflict registry entries, escalation response records, bounding verification records, perimeter integrity records, provenance annotations, amendment and review records, trust calibration evidence, and post-dissolution investigation records. These records exist not as a secondary audit trail but as the primary substrate content through which governance operates.

The consequence is that governance health is assessable from records alone. An observer with access to a Self’s governance records—without interviewing participants, without relying on declarations of compliance, without needing to observe the FAI events themselves—can determine whether each positive health indicator is satisfied and whether each warning indicator is present. This record-readability is the prior-art claim D2.35 establishes: governance health is objectively measurable through the record-keeping the architecture requires, and a governance health indicator that cannot be

assessed from records is not a governance health indicator in the CKS sense—it is a management opinion.

This premise has a practical implication for indicator design. Each indicator must be formulated as a question answerable from records. “Are configurations complete?” is record-answerable: the configuration substrate content either specifies all six dimensions or it does not. “Is the governance culture healthy?” is not record-answerable and is therefore not a governance health indicator in this framework.

2. Ten positive health indicators

The following ten indicators mark a governance portfolio in which the Phase D2 commitments are being honored. Each indicator is stated as a positive property and then specified as what the governance records show when the indicator is satisfied.

H1 — Configuration completeness. Every FAI event has an authored configuration substrate specifying all six dimensions established in D2.12. No dimension relies on an infrastructure default in place of an explicit governance decision. The governance records show: complete configuration entries for every event, with all six dimensions explicitly populated.

H2 — Timely joint authorization. Configuration authorization precedes construction by a governance-appropriate lead time. No event record shows construction commencing before joint authorization is complete, or authorization completed under compressed timelines that indicate structural bypass. The governance records show: authorization timestamps that precede construction records by the configured lead time, with no authorization entries dated after construction begins.

H3 — Conflict registry completeness. Every conflict detected during an FAI event—whether surfaced at merge, during exchange, or at dissolution—has a conflict registry entry carrying both sides of the conflict preserved, attribution to the contributing Selves, tier assignment (preserve, resolve, or escalate), and terminal status. No detected conflicts remain in indefinite open status. The governance records show: a conflict registry in which every entry has all required fields populated and every entry has reached a terminal state or has an active escalation record with response evidence.

H4 — Escalation responsiveness. Escalated conflicts receive governance responses within the configured response timeline established in D2.14. No escalation records show elapsed time exceeding the configured timeline without a substantive response entry. The governance records show: escalation records paired with response records, with elapsed times within configured bounds.

H5 — Exchange bounding verification. Exchange bounding verification records established in D2.16 exist for every FAI event. Verification is conducted rather than skipped, and verification records carry the specific bounding dimensions examined. The governance records show: a bounding verification entry for every event, with verification attributed to a governance actor and dated within the event’s governance window.

H6 — Home perimeter integrity confirmed. Post-event home perimeter integrity verification records established in D2.17 exist for every participating Self for every FAI event. Every Self that contributed aspects to the shared substrate has a record confirming that its home perimeter integrity was verified post-dissolution. The governance records show: integrity verification entries paired to each participating Self for each event, with no absent entries.

H7 — Evolution feed attribution. All FAI-origin records in home action layers carry FAI-origin provenance established in D2.10. No records in home substrates that trace to FAI events are present without attributing provenance to the originating FAI event and the contributing Selves. The governance records show: action-layer additions traceable to FAI events carrying complete provenance annotations with no gaps.

H8 — Standing configuration currency. Standing configurations have recent amendment records or recent review confirmation records within the configured review cadence established in D2.21. No standing configuration has an amendment or review record older than the configured cadence without a recorded currency confirmation. The governance records show: amendment or review entries for each standing configuration within the cadence window, with no stale configurations.

H9 — Trust calibration coherence. Sharing scope and provenance depth choices in event configurations are consistent with the governance compliance evidence accumulated from prior events, as established in D2.29. No event configuration shows expanded sharing scope or increased provenance depth without corresponding compliance evidence from the history of interactions with the relevant counterpart Selves. The governance records show: configuration entries whose trust parameters are within bounds justified by the compliance evidence the prior-event record supports.

H10 — Post-emergency investigation. Any emergency dissolutions in the governance portfolio have post-dissolution investigation records established in D2.28. No emergency dissolution is unexamined. The governance records show: an investigation record paired to every emergency dissolution entry, with the investigation record documenting the failure mode, contributing governance factors, and corrective commitments.

3. Ten warning indicators

The following ten indicators mark patterns that, if left unaddressed, drift toward the operational anti-patterns formalized in D2.19. Each warning indicator is a directional signal, not yet a confirmed failure, but pointing toward a named failure trajectory. Governance practitioners can detect each warning from records alone, without waiting for the failure to complete.

The mapping to D2.19 anti-patterns is given for each warning indicator. Two mappings are explicitly anchored in prior Phase D2 derivations: W3 drifts toward AP-3 (silent conflict collapse) and W5 drifts toward AP-8 (exchange bounding violation). The remaining mappings follow the same structural logic—each warning indicator is the early-stage record signature of a governance failure whose terminal form is named in D2.19.

W1 — Implicit configuration drift. An increasing frequency of FAI event dimensions defaulting to infrastructure settings rather than authored governance content. The governance records show: a trend of configuration entries with unspecified dimensions or explicit references to infrastructure defaults, increasing in frequency across the recent event portfolio. This drifts toward AP-1 (ungoverned configuration), in which FAI event parameters are determined by infrastructure rather than by human-authored governance decisions, and the architecture's configuration-as-substrate-content commitment is vacated in practice even while the structural form of governance is maintained.

W2 — Authorization lag. Joint authorization is frequently completed just before or after construction rather than with appropriate governance lead time. The governance records show: authorization timestamps and construction-commencement records with diminishing elapsed time between them, approaching zero or crossing to post-commencement authorization. This drifts toward AP-2 (authorization collapse), in which construction proceeds without substantive joint authorization, reducing authorization to a formal timestamp rather than a genuine governance decision.

W3 — Conflict registry gaps. Conflicts detected at merge have missing registry entries, one-sided entries with only one contributing Self's position preserved, or entries with absent tier assignments. The governance records show: conflict detection signals in event records that are not paired with complete registry entries, or entries that carry only one side of the preserved conflict. This drifts toward AP-3 (silent conflict collapse), in which conflicts are detected but not governed—the architectural default of preservation is bypassed and governance loses visibility into the inter-Self disagreements the shared substrate was constructed to surface.

W4 — Escalation backlog. Escalated conflicts are accumulating without governance responses, and response timelines configured in D2.14 are regularly exceeded. The governance records show: escalation entries with elapsed times exceeding configured response timelines and no substantive response entries, across multiple events in the recent portfolio. This drifts toward AP-4 (escalation abandonment), in which the third tier of the three-tier conflict mechanism becomes non-functional as a governance channel, and unresolvable inter-Self conflicts accumulate without human authority engaging them.

W5 — Bounding verification skipped. Exchange bounding verification records are absent for recent FAI events. The governance records show: a pattern of FAI events without corresponding bounding verification entries, indicating the D2.16 verification step is being omitted. This drifts toward AP-8 (exchange bounding violation), in which the instinct-layer protection that exchange bounding enforces is no longer verified, creating conditions under which FAI outputs could influence instinct evolution in violation of Paper 2's instinct-takes-no-FAI-input commitment.

W6 — Home perimeter unverified. Home perimeter integrity verification records are absent for participating Selves post-event. The governance records show: FAI dissolution records without paired perimeter integrity verification entries for one or more participating Selves, increasing in frequency. This drifts toward AP-6 (ungoverned home perimeter), in which the boundary that each

Self's home governance maintains against the shared substrate is no longer systematically verified, allowing boundary degradation to accumulate undetected.

W7 — Provenance decay. FAI-origin records in home substrates carry incomplete or absent FAI-origin provenance. The governance records show: action-layer additions traceable to FAI events that lack complete provenance annotations, with the gap between FAI participation and annotated home-substrate records widening. This drifts toward AP-7 (ungoverned inter-Self learning), in which the attribution chain linking home-substrate evolution to inter-Self exchange is broken, and the governance visibility over which Self's aspects influenced which evolution outcomes is lost.

W8 — Stale standing configurations. Standing configurations have no amendments or reviews within the configured cadence established in D2.21. The governance records show: standing configuration entries whose most recent amendment or review record is older than the configured cadence, with no currency confirmation entries. This drifts toward AP-5 (standing configuration staleness), in which the governance parameters that govern ongoing inter-Self relationships no longer reflect current governance intent, and the gap between authored configuration and operating reality widens without the record making the gap visible.

W9 — Trust inflation pattern. Sharing scope and provenance depth are increasing across successive events with the same counterpart Selves without corresponding growth in governance compliance evidence. The governance records show: event configuration entries with expanding trust parameters against a compliance evidence record that has not grown proportionally. This drifts toward AP-9 (trust inflation), in which governance-calibrated trust is replaced by relationship-habituated trust—each event extends trust because the prior event did, without the prior-event compliance evidence justifying the extension.

W10 — Unexamined failures. Emergency dissolutions have occurred without post-dissolution investigation records. The governance records show: emergency dissolution entries without paired investigation records, leaving governance failures unexamined and their contributing factors unrecorded. This drifts toward AP-10 (governance failure amnesia), in which emergency governance failures become invisible in the governance record, the portfolio accumulates unexplained dissolution events, and the corrective mechanisms that post-investigation records are designed to trigger are never activated.

4. The governance health dashboard

The twenty indicators can be organized as a governance health dashboard: a structured periodic review that governance practitioners conduct to assess the health of their FAI governance portfolio at the cadence specified in the network participation configuration per D2.31.

The dashboard operates as follows. At each review period, the governance practitioner assesses all twenty indicators against the governance records for the portfolio period under review. The ten positive indicators are assessed for presence: does the governance record show, for every FAI event in the review

period, that each indicator is satisfied? The ten warning indicators are assessed for absence: does the governance record show any of the warning patterns in the review period's event history?

The dashboard produces a structured finding rather than a score. Each positive indicator assessment results in one of three findings: satisfied across all events, satisfied with exceptions (specific events where the indicator was not met), or not assessable (records do not support a determination). Each warning indicator assessment results in one of two findings: absent (the warning pattern is not present in the review period's records) or present (the warning pattern is detectable, with the specific record evidence noted). A warning-present finding carries a trajectory designation: which AP-n anti-pattern the warning is drifting toward, based on the D2.19 mapping.

The dashboard does not produce a composite governance health score. Governance health is not scalar. A portfolio with nine of ten positive indicators satisfied and one warning indicator present is not "90% healthy"—it has a specific open risk (the drifting warning) against a specific anti-pattern trajectory, and the governance response is specific to that risk rather than to a composite assessment.

The dashboard's utility is portfolio-level: it makes visible the distribution of governance health across multiple concurrent FAI events and standing relationships, without requiring event-by-event manual review of every record. The twenty indicators serve as the review structure that compresses the governance record into a periodic governance conversation.

5. Why record-readability matters

The record-readability of all twenty indicators is not an implementation convenience. It is an architectural claim about what governance health means.

Governance health assessments that depend on participant self-reporting are assessments of participant beliefs about governance quality, not assessments of governance quality itself. A governance portfolio in which participants report high confidence in conflict handling may have a conflict registry with systematic gaps. A governance portfolio whose participants report conservative trust calibration may have configuration entries showing consistent trust inflation. The self-report and the record diverge whenever the governance practices participants intend differ from the governance practices the records show.

By requiring all twenty indicators to be assessable from records alone, D2.35 makes governance health a property of what governance practitioners actually did—what records they created, what entries they completed, what verifications they conducted—rather than a property of what they believe or intend. The distinction is the same distinction that makes a substrate the source of truth in the CKS architecture rather than participants' memories of what the substrate carries.

This does not eliminate the role of governance judgment. The governance health dashboard requires practitioners to assess records and reach findings. But the practitioner's judgment operates over substrate content—the records—rather than over self-reported assessments. The records constrain the judgment; they do not replace it.

The framework also permits third-party assessment. Because all twenty indicators are record-based, a governance auditor, a counterpart Self conducting trust calibration, or a population-level governance body can evaluate a Self's FAI governance health from records without relying on the assessed Self's cooperation beyond record access. This supports the calibrated-trust accumulation that D2.29 formalizes: a counterpart Self can treat governance compliance evidence as grounded in records, not as declarations.

6. Operational test

The governance health dashboard is operational for a given Self's FAI governance portfolio if and only if all of the following are true at the time of dashboard assessment:

1. The governance record contains a configuration entry for every FAI event in the review period, with all six dimensions specified.
2. Every conflict detected in the review period's events has a conflict registry entry with both sides preserved, attribution, tier assignment, and terminal status or active escalation record.
3. A bounding verification record and a home perimeter integrity verification record exist for every participating Self for every event in the review period.
4. Every FAI-origin addition to home action layers within the review period carries FAI-origin provenance.
5. Every standing configuration has an amendment or review record within the configured cadence as of the dashboard assessment date.
6. Every emergency dissolution in the portfolio history has a post-dissolution investigation record.
7. The compliance evidence record for each counterpart Self is sufficient to support the trust parameter choices in the most recent configuration for that counterpart.
8. Every escalation in the review period has a response record within the configured timeline.

A governance portfolio that satisfies all eight conditions has the records necessary to assess all twenty indicators. A portfolio that fails any condition has at least one indicator that is either unsatisfied (positive indicator) or affirmatively present (warning indicator). The dashboard assessment can proceed from the governance records without additional information from participants.

7. Conclusion

FAI governance health is observable. The Phase D2 governance record architecture was designed so that every commitment the FAI architecture makes produces a durable record, and those records collectively contain the signals that indicate whether governance is functioning or degrading. D2.35 formalizes twenty of those signals—ten that mark a healthy governance portfolio and ten that mark drift toward named failure modes—and organizes them as a governance health dashboard for periodic portfolio review.

The prior-art claim is this: a governance architecture in which health is not objectively assessable from records alone is a governance architecture that depends on declarations of compliance rather than evidence of compliance. The twenty indicators formalized here are evidence-based, record-grounded, and independently assessable. Governance practitioners who maintain the records the Phase D2 architecture requires have, as a direct consequence, the substrate from which their governance health is readable.

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